GE Medical Systems

Technical Publication

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IServ – GRE Console

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REVISION HISTORY

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Revision	Date	Reason for change
0	06/17/03	Initial Release

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ISERV – GRE CONSOLE

PROGRAM TEAM

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CA-GP (Customer Advocacy – Global Parts) Matt Parker

LSD (Lead System Designer) Parimal Shah

LSI (Lead Service Integrator) Nicole Boyle

LSSD (Lead Service System Designer) George Farrington

SME (Service Methods Engineer) George Farrington

1 PROGRAM OVERVIEW

The GRE Console program is focused on the introduction of the "Global Recon Engine Console" into the CT product line. The introduction of this console will be phased into the LightSpeed family of products. This program will address the first phase of this introduction. This first phase will deliver a PC (Linux) based console with a new recon engine that will cut into forward production of the LightSpeed 16 products. Variable Cost Productivity (VCP) and Serviceability are the two key driving factors of this new console. See Table 1-1 for the key program deliverables that this program will bring to the CT business.

TABLE 1-1
LIST OF DELIVERABLES FOR LS2002

IMPROVE ICV	Maintain/Improve Serviceability	NEW APPLICATIONS	NEW HARDWARE
$10K Reduction of Console ICV (US $)	Diagnostic parity with LightSpeed Plus 16	Module Diagnostics	Host PC
	System Calibration Time <3 hours	DIP Diagnostics	DARC
	Load From Cold Time Approx 45 minutes	Information Wizard Repair Wizard	SDDA
	DRFT / FRFT (Host PC 100% Coverage)	H/W & S/W Configuration Log	IG

2 PRODUCT SERVICEABILITY STRATEGY (M1.2.3.1)

This program will deliver serviceability in the form of reduced FRU count, increased Diagnose Right First Time (DRFT) and Fix Right First Time (FRFT) coverage. The bulk of these improvements will come in the form of a new Host Computer and recon engine modules. The Host Computer and recon engine modules will be a single high level repairable FRU. There will be no changes to the remaining "legacy" components.

A Hardware Configuration Log will incorporate all of the critical model and serial numbers of the system components in electronic form per the configuration tracker position paper.

Additional Non-Modality programs will be incorporated into this new product for improved serviceability. These will include:

- ILinq Generation II
- IIP 3.0
- eFlextrial
- Broadband Connectivity
- ScanPath
- ICenter

3 SERVICE METRICS DEFINED (M1.2.3.2)

Note: The Service Six Sigma web site that has Uptime by Modality and PSI Code (in Excel you need to Unhide rows to see by PSI Code).

1. Go to: http://3.45.92.210/sixsigma/.
2. Click on "Champions Handbook" at the top, then click on "Reporting Mechanisms".
3. There you will see "America's Uptime Reports", so click on it.

The key system service level metrics tracked for this program will be install (parts and labor), warranty (parts and labor), and PM (hours). The key Service Metrics tracked for this program will be MTBCC (Mean Time Between Customer Call), ETTR (Elapse Time To Repair), and % Remote Fix (OLC). The lead service integrator will track the metrics and report issues to the team. The specific goals for the service metrics are listed in Table 3-1.

The Service Metrics (MTBCC, WTTR, ETTR, PM, % Remote Fix) will be measured utilizing the GECARES database as well as pilot feedback QMI calls conducted during the M3 phase of the program.

TABLE 3-1
GRE SERVICE METRICS (GEMS-AM)

SUBSYTEM SERVICE GOALS	**GRE (@M3)**	**GRE (@M4)**
MTBCC (B10) Days MTBCC (B63) Days	10 300	20 300
Mean – WTTR (Onsite)	1	1
ETTR (P95-Hours)	18	18
ETTR (P75-Hours)	10	10
Planned Maintenance (Hours per year)	4	4
Installation Hours Average Hours per Installation per Site Average Cost per Installation per Site	1 mechanical 1 service	1 mechanical 1 service
Warranty Cost (Hours / Material Cost-US $)	15/$475	13/$410

4 DELIVERY AND PRODUCT INSTALLATION STRATEGY (M1.2.3.3 & M1.14.1.2)

All packaged components shall comply with the Global Packaging Guideline 2100268PRE. Manufacturing shall provide a means to protect the completed console assembly from weather and damage that could occur during handling and transportation for both domestic and international shipments. GRE will use a global shipping skid.

TABLE 4-1
CONSOLE ASSEMBLY CONFIGURATION PRIOR TO SHIPMENT

COMPONENT	OTHER ACTIONS
Console Base	Shipped on Global Shipping Skid
Console Software	Base software configuration installed
System Calibration Files	Provided from manufacturing
Software Options	Provided from manufacturing for installation at customer site
Monitor table top	Shipped installed
Key Board table top	Shipped separate from console for customer installation at site
Monitors	Shipped as separate Catalog item

Since Service prefers to outsource the mechanical aspects of installations and upgrades in some geographic locations, outsourced installation of mechanical tasks must be supported. Requirements are listed in Installation Document Book 1.

4-1 ANNUAL WARRANTY COSTS AT M4, M5 VS. CURRENT PRODUCT (M1.2.3.4.1)

See table 3-1 for estimated cost information.

Warranty cost reduction will primarily be a result of the new design and FRU philosophy. This is seen in the use of PC technology, and the new recon modules. The Host PC, DARC, SDDA, and IG nodes are PC based hardware sets. This results in industry standard components with higher reliability and lower cost. The reduction in cost allows the move to a higher level FRU for all major components. In turn this results in higher confidence in fault isolation and faster repair time.

4-2 ESTIMATED COST OF CORRECTIVE MAINTENANCE OF PILOT UNITS BETWEEN M3 & M4 (M1.2.3.4.2)

See table 3-1 for estimated cost information. Corrective Maintenance cost reductions rely heavily on the ability to rapidly diagnose and repair problems in the Console. This will require full implementation of module isolation, reli monitoring, and repairable PC and modules (globally). New technology in the Host PC and modules will improve the ability to perform a Load From Cold of the software. This will reduce the time for Host PC replacement as well as future software upgrades.

5 SERVICEABILITY REVIEW PLAN (M1.9.8.1)

The serviceability reviews will correspond to the design reviews for the following:

- Software Installation
- Diagnostics Coverage (On-Site and Remote)
- Hardware Repairability (meeting with vendor and GPO)
- General Overview
- Recon Overview
- Service Review
- FMEA for DRFT and FRFT at all module levels

As the Image Reconstruction Chassis has been separated into modules (Host PC, DARC, IG, and SDDA) into a single box, a review of the diagnostic coverage must be completed to assure DRFT and FRFT capability at this level.

5-1 DIAGNOSTICS, CALIBRATION, AND IMAGE QUALITY TOOLS (M1.9.8.1.1)

The existing calibration and image quality tools in the H16 program are scheduled to be ported to the new Linux software platform in their entirety. Additions to this tool set will include:

- Dip Diagnostics – equivalent to or better than those existing on LightSpeed Plus
- PC and module diagnostics for the Host PC to provide 100% diagnosability to the FRU

5-2 ERROR MESSAGE ANALYSIS (M1.9.8.1.2)

The introduction of a PC and modules into the Console may result in new error messages. Additional environmental monitoring will also result in new error messages. These messages must be reviewed for completeness and accuracy. This review will be incorporated into the “Diagnostic Coverage” review.

5-3 INSTALLATION PROCESS (M1.9.8.1.3)

This program will have a minimum impact on the installation process as compared to the current console. The new console is designed to have the same footprint of the original console.

5-4 SERVICE METRICS PLAN FOR M3, M4, M5 (M1.9.8.1.4)

The primary service metric that will be impacted by this program is the warranty cost. The opportunities for this cost reduction are outlined in section 4-2.

5-5 SERVICE TOOLS / DIAGNOSTICS PACKAGING AND PROPRIETARY CLASSIFICATION PLAN (M1.9.8.1.5)

This program will impact the following Diagnostics tools:

- Class C Software Tools (distributed on CD ROM)
- Class M Software Tools (distributed on CD ROM)

A new Advanced System Service Manual specifically for the GRE will be created, and it will incorporate the new console hardware, diagnostics, and repair procedures.

6 SERVICE DELIVERY INTEGRATION (M1.9.8.2)

This section describes the plans for OLC/field training, serviceability reviews, site support during clinical, pilot and production phases of the program, and the feedback process during these phases.

6-1 TRAINING/SUPPORT PLAN FOR ON LINE CENTER (OLC) AND FIELD ENGINEER (FE) (M1.9.8.2.1)

A member of the GEMS-AM OLC team will be assigned as a member of the program core team. This individual will be closely involved in the review and design discussions. This process will assure an "expert" status for that individual. This expertise will then be the focal point for ongoing training of the OLC engineers globally. As often as possible, the local (GEMS-AM) OLC engineers will be incorporated in the system testing to increase hands-on experience.

Beyond the above education process, there will be ERT (Early Readiness Training) sessions held to provide training on the new system. The first training overviews will be incorporated in the 2003 CT Service Seminar held FW-13 in Milwaukee.

6-1-1 EXTERNAL EVALUATION (EE) ON-SITE SERVICE SUPPORT PLAN (M1.14.2.2)

The EE sites will be supported through a standard escalation process with the engineering design team ultimately owning the support of the systems. With this plan comes the need for custom training of the primary field engineers for each of the EE sites. This will be accomplished via on-site and remote engineering support. As the EE sites will be installed prior to completion of full training packages, the primary field engineer will be given a list of contact names and phone numbers to assure rapid site failure identification and resolution. A weekly QMI call will be held to communicate site status and upgrade plans between the engineering team and the field team. This call will be lead by the LSI of the program.

6-1-2 PRE-PRODUCTION (PILOT-M3) SYSTEMS

The Pilot plan includes the shipment of approximately 150 systems in Q3 of 2003. At the release of the pilot systems, all service documentation will be complete and capable of supporting the product for service features available. As the OLC engineers will be participating in the EE support, they will be asked to identify special training needs required to enable full support by the OLC at the M3 of this product. The use of the OLC engineers in the reviews and testing of the designs will facilitate this.

ERT training sessions will be developed to provide the necessary education to support the installation and maintenance of the pilot systems. This training will be done as a joint effort between the engineering team and the education center. The deployment mechanism for this training will initially be a web based package that is discussed on T-Cons. This enables the interaction necessary to improve the early introduction learning curve. T-Con sessions will be scheduled on multiple dates and times to allow for global participation.

6-1-3 PRODUCTION (M4) SYSTEMS

By the release of the M4, the education center will be responsible for the ongoing education process to support this product. This will be accomplished through the use of remote training tools and resident education.

Service documentation and training will be updated based on the feedback from the pilot shipments. The QMI feedback will be continued for support of any upgrades of the pilot sites.

A cross-functional serviceability review will be held on a fully functional system with participants from Engineering, OnLine Center, Education Center, Field FE's, Installation Support Services, CA-GP, and Service EHS. Service representatives from GEMS-E and GEMS-A will be included as available. The review will be scheduled and completed between milestones M1 and M2. The program LSI will handle review logistics (time, place, and communication to participants). The Service Methods Engineers will work with the designers to identify presenters and specify/review content of their presentations. Design Engineers shall present material relating to the following topics:

- Diagnostics, Calibration, and Error Message Analysis
- Installation Process
- Service Tools / Diagnostics Packaging
- FRU Strategy
- PM & Corrective Maintenance Strategy

Review output will be forwarded to the applicable GEMS engineers and vendors.

6-2 SERVICE SYSTEMS / OLC INTEGRATION PLAN (M1.9.8.2.3)

The OnLine Center (OLC) must be prepared to support pilot systems beginning at M3. The Program Service Integrator will work with the OLC management to identify an acceptable training/integration plan for OLC engineers. The plan will include identification of a lead OLC engineer from each pole for the program, and a training plan for all OLE's who will be supporting pilot sites. The plan will include the roles/responsibilities of the lead OLC engineers. Desired roles/ responsibilities include: being available for some team meetings, being a communication conduit to the core team on program related issues, coordination of training at OLC, participating in pilot teleconference calls, and providing status of support calls during product rollout.

6-3 PROGRAM / SERVICE QMI (M1.9.8.2.4)

The program team is responsible to monitor clinical performance and collect necessary feedback from clinical sites throughout the program development cycle. Weekly calls will be used to inform the field about known program pilot issues and to obtain system feedback from the EE & M3 sites. The program LSI will be the Team Leader and organize the call logistics (who-what-when). The first teleconference call will be held after the installation begins on the first EE system; the calls will continue until M4. Participants on the call shall include (but is not limited to) pilot site FE's, Install Specialists, Lead OLC Engineers, program LPI, LSD, LSSD, SMEs, and other program team members as appropriate. Applicable issues will be entered into the CQA system (preferably by the person identifying the issue) for tracking and resolution.

7 PLANNED MAINTENANCE STRATEGY (M1.9.8.4)

The planned maintenance for this product will be reviewed and revised to utilize the philosophy of event based maintenance. This will eliminate some of the time based inspections and replace them with event based inspections. This will be accomplished by use of counters and sensors in the system.

8 CORRECTIVE MAINTENANCE STRATEGY (COSTS BEYOND M3 – M4) (M1.9.8.5)

The primary focus of this program is the replacement of the VME console with a Linux base hi-speed modular recon console. This program will follow the previous program's lead in making the Host computer and recon modules a single repairable FRU. Each host PC, SDDA, DARC, and IG module has a 3-year warranty. The field will use the SWAP program to exchange these parts.

As a vitality program, Service Marketing will introduce a contract package to provide a scheduled replacement of the Host Computer for long term contract purchasers. This will allow the customer to maintain current technology and spread the cost over a contract period equal to the life expectancy of the computer (3 years).

9 SERVICE DOCUMENTATION PLAN (M1.9.8.6)

Since the GRE will be used on multiple systems, there are many documents that will need to contain information about the new console. We will create one document for the GRE and include it with the H16 and 4/8 slice systems in forward production. The documentation sets impacted are detailed in Table 3-1. The Load From Cold Procedure will be on paper, but the rest of the service documentation will be on a CD.

TABLE 3-1
LIGHTSPEED 16 SLICE DOCUMENTATION LIST

DIRECTION	TITLE	CLASS	DISTRIBUTION
2384451-200	System Service Manual	A	CD
2384451-200	Advanced System Service Manual	C	CD
2378261-100	SW Load Procedure	A	Paper with system
2384451-200	Pre-Installation / Site Planning Manual	A	CD
2384451-200	Installation Manual	A	CD
2384451-200	Schematics Manual	A	CD
2384451-200	Illustrated Parts Manual	A	CD
2370773-100	Applications Tips & Workarounds	A	Paper with system
2351785-100	Technical Reference Manual	A	Paper with system
2369740-271	Learning Reference Guide	A	CD
2141834-100	CT PM PROCEDURES	A	CD

NOTE: A COPY OF THE INTERCONNECTS (ONE LARGE SCHEMATIC) WILL BE SHIPPED WITH THE CONSOLE. IT CAN BE FOUND ON THE BACK COVER OF THE CONSOLE.

10 FRU AND REPAIRABLE PART STRATEGY (M1.9.8.7)

Detailed in following sub-sections 10-1 & 10-2

10-1 FRU GUIDELINES

The following guidelines shall be used for design development, FRU identification and troubleshooting. Active type FRU's are assemblies that are powered and contain active devices such as circuit boards, power supplies etc. Passive type FRU's are parts that are not powered such as nuts, bolts, fiberglass pieces, chassis frames etc. FRU's for this program will be 1 or "No."

By design:

- All active type FRU's, whenever possible; must have the ability of remote isolation from the OnLine Center. These would be the Host computer, SDDA, IG, DARC, SCSI Tower, monitors, trackball, hardware, and cable kits. Passive type FRU's must be isolatable on site.
- New FRU's, whenever possible, should be backward compatible to the Installed Base product.

10-2 FRU AND REPAIRABLE PARTS PLAN

The Field Replaceable Unit (FRU) plan will address only the new FRU's added to the system as a result of the program. The following FRU identification and stocking process plan will be implemented and complete by M3.

The Program Service Integrator will coordinate a cross-functional FRU review of the new parts introduced by this program. Planned reviewers are Engineering, Service Engineering, CA-GP, Manufacturing, OnLine Center Engineer, Field Engineer and the Education Center. This review and product structure updates must be completed prior to zero release of any drawings / BOMS and M2.

The Service Integrator will coordinate a FRU stocking review for the new parts. Reviewers are Service Engineering, CA-GP, and Manufacturing. The FRU stocking plan must be completed at M2.

All new FRU's to support clinical sites will be provided by the program. New FRU's identified at the FRU review to be in stock, must be complete and on the shelf globally at M3.

New FRU1s costing > $2500 will be considered for repair (T3). Repair Operations shall be included in any discussions of repairable part plans.

10-3 NEW FRUS FOR THIS PROGRAM

TABLE 10-1
LIST OF NEW FRUS FOR GRE

FRU Description	Part Number	Repairable/Consumable
GRE Console (Complete)	2377708	Consumable
IG	2362872	SWAP
SDDA	2362873	SWAP
Front Console Cover	2369619	Consumable

DARC	2362870	SWAP
Cable Kit Bulkhead	2367805	Consumable
USB Trackball	2376560	Consumable
Fan Assembly	2376909	Consumable
Outlet Box Assembly	2174159-3	Consumable
GRE Hardware Kit	2379133	Consumable
Transceiver	2377227	Consumable
SCSI Cable Kit	2379129	Consumable
Power Switch and Cables	2379134	Consumable
Ethernet Communication	2379130	Consumable
GRE Console Slide Service Kit	2379135	Consumable
Video/Power Cable Kit	2379131	Consumable

NOTE: ALL FAILED PARTS WILL BE RETURNED TO CT ENGINEERING TO BE ANALYZED. WE WILL USE THE PARTS AND A PAGER PROCESS.

11 SERVICEABILITY FEASIBILITY DOCUMENTED (M1.12.4)

In general, service feasibility for the following will be verified round M0/M1:

- Diagnostics
- Service Tools
- Proprietary Classification for A/B and C Materials
- Installation Process
- Remote Access and Remote Serviceability

A Service Feasibility review will be conducted with the rest of the program team at completion of the clinical verification. Issues identified will be prioritized and plans put in place to reach resolution. For more detailed serviceability plans/results, see the following subsections.

11-1 DIAGNOSTICS (M1.12.4.1)

The Host PC and modules will introduce a new technology to the CT product line. The use of a PC based computer, modules, and Linux operating system will require a new approach for the diagnostics. The power up set of diagnostics will be reviewed for DRFT ability and ease of use during the ME cycle.

Introduction of the PC platform will also require a review of the error reporting by the system. This error reporting will be reviewed at ME for accuracy and consistency with the previous platform.

11-2 SERVICE TOOLS (CALIBRATION, IMAGE QUALITY, OTHER) (M1.12.4.2)

No new tools are scheduled to be introduced with this program other than those defined in section 11-1.

11-3 CONFIRM SERVICEABILITY PROPRIETARY CLASSIFICATION FOR A/B AND C MATERIALS (M1.12.4.3)

A review of the new diagnostics will be conducted to assure proper compliance with proprietary guidelines.

11-4 INSTALLATION (M1.12.4.4)

There are no changes currently scheduled that will impact the installation of the console. The new console should be simpler to install. CT Service Engineering will also include the schematics in the back of the console to assist the FE and make the install easier.

11-5 REMOTE ACCESS AND REMOTE SERVICEABILITY (M1.12.4.5)

This product will contain the same remote access as the current product supports. This includes modem and Broadband connectivity. Broadband will be standard. The modem will be a purchasable option.

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